

Question number	Answer	Additional guidance	Mark
4(a)(i)	The correct answer is D <i>A is incorrect because the correct answer is $(2.5 \times 10 \times 1000)/0.14 = 178\,571.42857 = 179\,000$</i> <i>B is incorrect because the correct answer is $(2.5 \times 10 \times 1000)/0.14 = 178\,571.42857 = 179\,000$</i> <i>C is incorrect because the correct answer is $(2.5 \times 10 \times 1000)/0.14 = 178\,571.42857 = 179\,000$</i>		(1)

Question number	Answer	Additional guidance	Mark
4(a)(ii)	An explanation that includes three of the following points: • (wall of) aorta has {elastic fibres / (smooth) muscle / collagen} in it (1) • because the blood is under high pressure (from the heart) (1) • capillary (wall) is only one cell thick (1) • because this is where {gas exchange takes place (in the lungs) / plasma needs to pass (out) / tissue fluid needs to pass (in)} (1)	ACCEPT (elastic fibres) for (elastic) recoil / to squeeze / pumps blood along / maintains high blood pressure ACCEPT oxygen passes in / carbon dioxide passes out (in lungs)	(3)

Question number	Answer	Additional guidance	Mark
4(b)	An answer that includes the following points: • (duration of) (ventricular) diastole and (ventricular) systole decrease with increase in heart rate (1) • manipulation of figures to illustrate one change in duration (1)	ACCEPT converse Piece together e.g. ventricular diastole decreases by 0.39 seconds / 74% ventricular systole decreases by 0.11 second / 40%	(2)

Question number	Answer	Additional guidance	Mark
4(c)	An explanation that includes three of the following points: • (peak) rate of flow in forward direction decreases with distance from heart OR (peak) rate of flow in backward direction increases with distance from heart (1) • rate of flow decreases (with distance from heart) due to less {force / pressure} (1) • due to (increased) {resistance (between blood and vessel walls) / branching (of aorta) / friction (between blood and vessel walls)} (1) • some backflow of blood {corresponding with diastole / no valves in aorta} (1)	ACCEPT description of both atrioventricular and semilunar valves being open description of both atria and ventricles {being relaxed / having low pressure} elastic recoil	(3)